LESSON 3.10

Convert Measurement Units

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can convert measurement units using ratios and conversion factors.

WHY THIS MATTERS You're building a brand-new arcade cabinet! The blueprints use feet, but the parts catalog lists measurements in inches. The wiring instructions use yards, and the weight limits are in pounds — but the shipping labels use ounces. You need to convert between units to make sure everything fits and ships correctly. The blueprint says the cabinet is 6 feet tall, and the parts catalog lists a side panel that is 70 inches tall. Does the panel fit?

REMEMBER

Convert Measurement Units

Dimensional Analysis & Unit Conversions A unit is a standard amount used to measure, like a foot or an inch. A conversion factor is a ratio that relates two units, like 1 foot = 12 inches.

1. Write the given measurement as a fraction over
2. Multiply by conversion ratios arranged so unwanted units cancel diagonally.
3. Multiply numerators, multiply denominators, and simplify the remaining unit

Words: **Conversion Factor** ***(Factor de conversión)*** · **Unit** ***(Unidad)*** · **Equivalent** ***(Equivalente)*** · **Ratio** ***(Razón)*** · **Customary units** ***(Unidades usuales (sistema inglés))***

1 **Circle the letter of the best answer. Show how you know.**

Which ratio is equivalent to 8:12?

(A) 2:3
(B) 4:8
(C) 8:16
(D) 12:8

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Fill in the missing value: 5:9 = 20:___

(A) 24
(B) 32
(C) 36
(D) 45

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

3 **Circle the letter of the best answer. Show how you know.**

A ratio table shows 7:4 and 21:12. What was each part of 7:4 multiplied by?

(A) 3
(B) 4
(C) 7
(D) 14

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

4 **Circle the letter of the best answer. Show how you know.**

Which pair of ratios is NOT equivalent?

(A) 3:7 and 9:21
(B) 4:6 and 10:15
(C) 5:8 and 15:25
(D) 6:10 and 9:15

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The arcade is ordering a new game cabinet. The doorway is 80 inches tall. The cabinet is listed as 7 feet. Your friend says it won't fit because 7 feet is 82 inches. How many inches are in 1 foot?

(A) 10
(B) 12
(C) 16
(D) 36

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

The arcade is ordering a new game cabinet. The doorway is 80 inches tall. The cabinet is listed as 7 feet. Your friend says it won't fit because 7 feet is 82 inches. How many inches tall is the 7-foot cabinet?

(A) 19
(B) 70
(C) 82
(D) 84

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do you decide whether to multiply or divide when converting units?

SENTENCE STARTER When converting from a ___ unit to a ___ unit, I ___ by the conversion factor because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Equivalent Measurements

_____ 2 gallons of coolant	(A) 8 quarts
_____ 60 inches of trim	(B) 5 feet
_____ 3,000 meters of cable	(C) 3 kilometers
_____ 32 ounces of bolts	(D) 2 pounds
_____ 10 feet of railing	(E) 120 inches

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

The arcade is ordering a new game cabinet. The doorway is 80 inches tall. The cabinet is listed as 7 feet. Your friend says it won't fit because 7 feet is 82 inches. Will the cabinet fit through the 80-inch doorway?

(A) No, because 84 > 80
(B) Yes, because 84 < 80
(C) Yes, because 7 feet is small
(D) You cannot tell without the width

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Leg 1: The joystick rod is 3 feet long, but the parts catalog lists it in inches. How many inches is that? (1 ft = 12 in)

(A) 15 inches
(B) 30 inches
(C) 33 inches
(D) 36 inches

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Leg 2: A power cable on the cabinet plan measures 2.5 kilometers across the whole arcade network. How many meters is that? (1 km = 1,000 m)

(A) 25 meters
(B) 250 meters
(C) 2,500 meters
(D) 25,000 meters

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Leg 3: The coin box weighs 5 pounds, but the shipping label needs the weight in ounces. How many ounces is that? (1 lb = 16 oz)

(A) 16 ounces
(B) 21 ounces
(C) 75 ounces
(D) 80 ounces

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — Multiply or Divide?

1. **Convert 4 feet of side panel to inches** 4 feet → inches
2. **Use the conversion factor** 1 foot = 12 inches
3. **Operate** 4 ÷ 12 = 0.33 inches
4. **Final answer** 4 feet ≈ 0.33 inches

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do you decide whether to multiply or divide when converting units?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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